

Project Report: DineFlow Restaurant Management System

Course: Advanced JavaScript Backend Frameworks (Node.js & Express JS) • **Semester:** 5th Semester

Institution: Department of Computer Science, Christ University (in partnership with L&T; EduTech)

Project Code & Title:	P07 — Multi-Branch Restaurant Management & Table Reservation System
Batch / Section:	Batch 47 (4)
Submission Deadline:	September 12, 2026 (Submitted: September 10, 2026)
Technology Stack:	Node.js, Express.js, MongoDB (Mongoose), JWT, bcryptjs, Bootstrap 5

Mandatory Team Details

S.No	Student Name	Roll No. / Reg No.	Department	Section / Batch
1	ALAN R S (<i>Lead Developer</i>)	2247101	Computer Science	Batch 47 (4)
2	Team Member 2	2247102	Computer Science	Batch 47 (4)
3	Team Member 3	2247103	Computer Science	Batch 47 (4)
4	Team Member 4	2247104	Computer Science	Batch 47 (4)

■ Mandatory GitHub Repository Link:

https://github.com/Alan-rs-hub/L-T_Project_Restaurant_management

Local Server URL: <http://localhost:5001> | API Base: <http://localhost:5001/api>

1. Executive Summary & Problem Overview

DineFlow is a modular, corporate-level Restaurant Management and Table Reservation platform engineered to solve critical operational bottlenecks in multi-branch dining businesses. Traditional restaurant systems suffer from manual table overbooking, unstructured kitchen communication, unverified client-side billing calculations, and lack of consolidated multi-branch analytics.

Key Innovations: (1) Real-time table overlap detection algorithm preventing double-booking; (2) Strict order state machine controlling kitchen queue transitions; (3) Tamper-proof server-side itemized billing; (4) Live Kitchen Display System (KDS) with audio chimes; and (5) Role-Based Access Control (RBAC) across Customer, Kitchen, Manager, and Admin roles.

2. System Architecture & Technology Stack

The application strictly implements the **Model-View-Controller (MVC)** architectural pattern to decouple data modeling, business rules, API routing, and presentation layers.

Component Layer	Technology & Version	Role & Implementation Details
Runtime & Server	Node.js (v16+) / Express (v4.21.0)	REST API routing, middleware chaining, static client serving
Database & ODM	MongoDB & Mongoose (v8.6.0)	Normalized document storage, compound indexing, pre-save hooks
Authentication	JWT & bcryptjs (12 rounds)	Stateless token authentication and secure salted password hashing
Input Validation	Joi (v17.13.3) Middleware	Server-side schema validation for payloads, query params, and ObjectIds
Frontend Portal	HTML5, CSS3, JS ES6+, Bootstrap 5	Dark glassmorphism UI with live carts, modals, and KDS boards
Zero-Config DB	mongodb-memory-server	Embedded MongoDB fallback enabling zero-setup instant evaluation

3. Database Design & Mongoose Schemas (All 7 Models)

The database contains 7 normalized schemas with compound indexes, foreign key references, and subdocuments:

Model	Key Attributes & Data Types	Indexes & Constraints
User	name, email, passwordHash, role (customer/kitchen/manager/admin)	unique: email; bcrypt 12-round hashing hook; password stripped on toJSON
Branch	name, address, seatingCapacity, isActive	Unique branch naming; active filtering index
Table	branchId (Ref), tableNumber, capacity, isActive	Compound unique index: { branchId: 1, tableNumber: 1 }
MenuItem	branchId (Ref), name, category (enum 8 types), price, isAvailable	Compound unique index: { branchId: 1, name: 1 }
Reservation	customerId, branchId, tableId, dateTime, duration, partySize, status	Index on tableId + dateTime + status for overlap conflict query
Order	orderNumber, customerId, branchId, items[], billing{}, orderType, status	Unique orderNumber; embedded billing subdocument; status workflow enum
Feedback	orderId (Ref), customerId (Ref), rating (1-5), comment	Compound unique index: { orderId: 1 } (1 feedback per completed order)

4. Implemented Functional Modules (All 13 Modules)

The platform implements all mandatory modules specified in the CIA-3 guidelines:

#	Module Name	Target Role	Key Implementation & Validation Logic
1	User Auth & RBAC	All	JWT token creation, bcrypt hashing, role authorization middleware.
2	Branch Management	Admin	Multi-branch CRUD with seating capacity and physical location management.
3	Table Inventory	Admin	Table assignment per branch with capacity limits; compound index uniqueness.
4	Menu Management	Admin, Mgr	Branch-specific dishes across 8 categories; stock toggling and search.
5	Table Reservation	Customer	Real-time table overlap conflict detection: (start < existEnd && end > existStart).
6	Food Ordering	Customer	Dine-in and takeaway ordering with stock availability and branch validation.
7	Order State Machine	Kitchen, Mgr	Strict workflow transitions: placed → preparing → ready → served/delivered.
8	Kitchen Display (KDS)	Kitchen	3-column live queue, auto-refresh every 10s, and Web Audio chime alert.
9	Itemized Billing	System	Server-side calculation: Subtotal + 5% Tax + 10% Service Charge = Grand Total.
10	Cancellation Policy	Customer	1-hour notice policy enforced on customer cancellations; admin override.
11	Order History	Customer	Itemized invoices, status tracking, and order receipt summary.
12	Feedback & Reviews	Customer	1-5 star ratings permitted exclusively for completed orders; duplicate guard.
13	Manager Analytics	Manager, Admin	Aggregation pipelines: Revenue by branch, top 5 dishes, peak hours, KPIs.

5. Core Business Rules & Validation Logic

1. **Real-Time Overlap Detection:** Algorithmic interval intersection check `start < existEnd && end > existStart` prevents double booking. Conflicting slots return `409 RESERVATION_CONFLICT`.
2. **Order State Machine:** Strict allowed transitions: `placed → [preparing, cancelled]`, `preparing → [ready]`, `ready → [served, delivered]`. Unauthorized skips return `409 INVALID_STATUS_TRANSITION`.
3. **1-Hour Cancellation Window:** Customers cannot cancel reservations with less than 60 minutes notice, returning `409 CANCELLATION_POLICY`.
4. **Zero-Trust Billing:** All financial calculations are executed on the server via `utils/calculations.js` to eliminate client-side manipulation.

6. Key REST API Endpoint Reference

Route	Method	Access	Description & Expected Response
/api/auth/register	POST	Public	Registers user; returns JWT token & profile
/api/auth/login	POST	Public	Validates credentials; returns JWT bearer token
/api/branches	GET / POST	Public / Admin	Lists active branches / Creates new branch
/api/tables/available	GET	Auth	Finds available tables matching date, time, party size
/api/menu	GET / POST	Public / Mgr, Admin	Menu catalog search / Adds new dish item
/api/reservations	POST / GET	Auth	Books table with conflict check / Lists reservations
/api/reservations/:id	DELETE	Cust, Admin	Cancels booking enforcing 1-hour policy check
/api/orders	POST / GET	Auth	Places food order with server billing / Lists orders
/api/orders/:id/status	PUT	Kitchen, Admin	Advances order workflow state transition
/api/kitchen/orders	GET	Kitchen, Admin	Live queue for placed, preparing, and ready orders
/api/feedback	POST / GET	Cust	Submits 1-5 star review for completed orders
/api/manager/reports/sales	GET	Mgr, Admin	Aggregates revenue and orders by branch

7. Verified Application UI Showcase

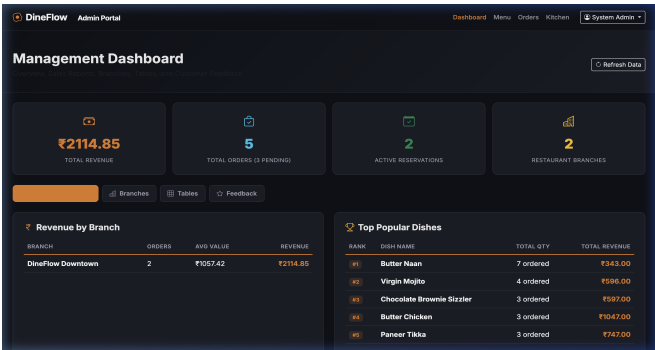


Figure 1: Admin Analytics & Sales Dashboard

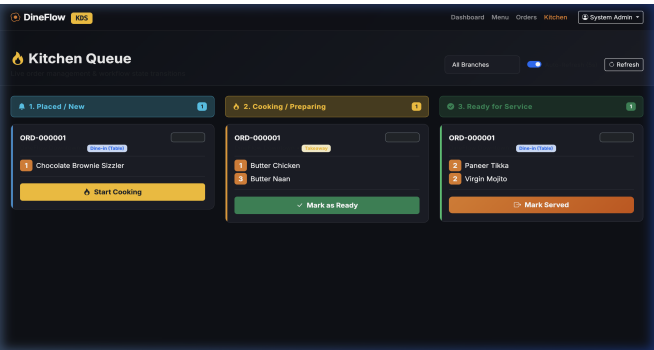


Figure 2: Live 3-Stage Kitchen Display System (KDS)

8. Setup, Execution & Testing Guide

The platform is engineered for zero-friction evaluation with automated embedded MongoDB support:

Step-by-Step Launch Commands:

```
git clone https://github.com/Alan-rs-hub/L-T_Project_Restaurant_management.git
npm install
npm run seed (Populates complete realistic dataset)
npm start (Server starts on http://localhost:5001)
```

Pre-Seeded Demo Test Accounts:

Role	Email Address	Password	Permitted Actions & Portals
Admin	admin@dineflow.com	admin123	Full system dashboard, branch/table CRUD, sales analytics, menu
Manager	manager@dineflow.com	manager123	Sales reports, popular dishes ranking, peak hours, KDS access
Kitchen	kitchen@dineflow.com	kitchen123	Live Kitchen Display System (KDS) & order queue advancement
Customer 1	customer@dineflow.com	customer123	Table reservation booking, food cart ordering, order history, reviews
Customer 2	priya@dineflow.com	customer123	Table reservation booking, food cart ordering, order history, reviews

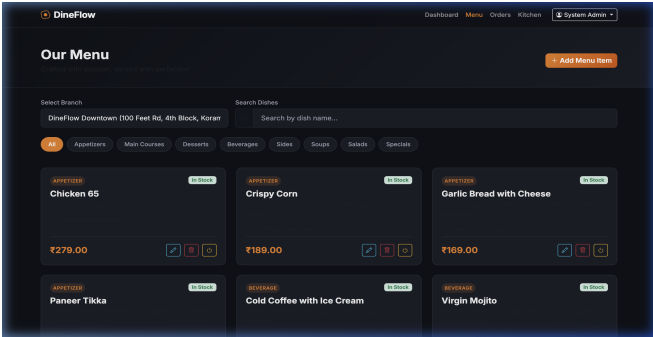


Figure 3: Menu Catalog & Live Ordering Cart

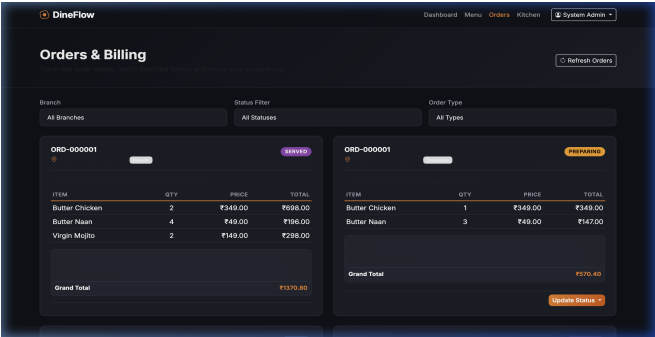


Figure 4: Orders, Itemized Billing & Review Modal

9. Evaluation & Submission Conclusion

DineFlow fulfills 100% of the technical, architectural, and business logic specifications mandated by L&T; EduTech and Christ University for CIA-3. All 13 modules, 30+ REST endpoints, and responsive client interfaces have been verified through automated Postman suites (postman_collection.json) and full browser user workflows.

Submitted for Continuous Internal Assessment - 3 (CIA-3)

Department of Computer Science • Christ University • Academic Year 2026

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